

Skew partitions, continued fractions, and connected components

A complete proof of the generating function
recorded as conjectural in OEIS A225114

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Abstract

Let a_n count skew partition diagrams with n cells and no empty rows or columns, including one empty diagram. We prove that

$$\sum_{n \geq 0} a_n q^n = \frac{1}{2 - C(q)}, \quad C(q) = \frac{1}{1 - \frac{q}{1 - \frac{q}{1 - \frac{q^2}{1 - \frac{q^2}{1 - \ddots}}}}},$$

where every power q^j occurs twice. The proof gives explicit bijections from row-overlap data to diagrams, from connected diagrams to pairs of boundary paths, and from those pairs to weighted two-colored Motzkin excursions. It also establishes refinements by area, rows, columns, and connected components; a closed formula at fixed dimensions; an independent ratio of q -series; a sharp square-law truncation bound; and exponential asymptotics with a rationally certified dominant-singularity interval.

The target was selected by one recorded random draw from a frozen shortlist. Although OEIS still describes its generating function as conjectural, Dima Pasechnik published a proof sketch in September 2024. Accordingly, this is a self-contained proof and elaboration, not a claim to the first solution of an unresolved conjecture. Exact independent checks through area 1000 and all source code accompany the article.

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1 Selection, provenance, and the main result

1.1 The frozen candidate list

The search was restricted to explicit conjectured ordinary generating functions for natural combinatorial or sequence-theoretic objects. Prime number and congruence questions were excluded. The following four candidates were fixed before a random index was generated. This is a screened shortlist, not a claim to survey every eligible OEIS conjecture.

Index	Entry	Subject	Conjecture attribution
0	A225114	Skew partitions without empty rows or columns	Kurkov, 2024
1	A244475	Fifth-largest distinct value in a row of Stern’s triangle	Heinz, 2022
2	A381190	Connected minimal dominating sets of trapezohedral graphs	Arndt, 2026
3	A391632	Mutual-visibility sets of Andrásfai graphs	Howroyd, 2026

Table 1: The ordered list used for the draw. The entries and their explicit formulas were checked on OEIS [1, 2, 3, 4]. The three unselected formulas are reproduced in Appendix A.

The recorded method was `random.Random(seed).randrange(4)`, with a 128-bit seed obtained from `secrets.randbits(128)`. The seed was

203225573310603175027560032786773102290.

The result was the zero-based index 0, hence **A225114**. There was one draw and no reroll. The files `candidates.json` and `selection.json` preserve the list and result; the verification script checks the list’s digest and replays the draw. This records the actual selection procedure, not merely a seed chosen after selecting a target.

1.2 What is being proved, and what was already known

The selected entry counts skew partitions whose diagrams have no empty rows or columns. It attributes the conjectured continued fraction to Mikhail Kurkov, 3 September 2024 [1]. These diagrams connect a familiar partition-theoretic construction with lattice-path and polyomino enumeration; they are not an artificially defined numerical sequence.

A subsequent prior-art check found Pasechnik’s MathOverflow answer of 10 September 2024 [5]. That answer sketches the generating-function identification and the ordered-sequence construction. The connected-diagram continued fraction is older: it appears in the record for A006958 and in work on skew Ferrers diagrams [6, 7]. The present article therefore retains the randomly selected target, proves its formula in full, and supplies additional derivations and consequences. It does not assert novelty of the connected continued fraction, the sequence construction, or the underlying enumerative identities.

Theorem 1.1 (The selected generating function). *Let a_n be the number of normalized skew partition diagrams of area n with no empty rows or columns, with $a_0 = 1$. Define*

$$C(q) = \frac{1}{1 - \frac{q}{1 - \frac{q}{1 - \frac{q^2}{1 - \frac{q^2}{1 - \frac{q^3}{1 - \frac{q^3}{1 - \ddots}}}}}}}. \quad (1.1)$$

The continued fraction is well defined coefficientwise in $\mathbb{Z}[[q]]$, and

$$A(q) := \sum_{n \geq 0} a_n q^n = \frac{1}{2 - C(q)}. \quad (1.2)$$

The essential decomposition is

$$\boxed{A(q) = \frac{1}{1 - P(q)}, \quad C(q) = 1 + P(q),} \quad (1.3)$$

where $P(q)$ counts nonempty *edge-connected* diagrams. Neither equality will be inferred from a finite list of coefficients. Sections 2 and 3 prove the geometric constructions, and Section 4 proves the continued-fraction identity.

2 Row lengths, overlaps, and connected components

2.1 Normalization and the exact counting convention

Definition 2.1. A nonempty normalized skew partition diagram is specified by partitions $\lambda = (\lambda_1, \dots, \lambda_h)$ and $\mu = (\mu_1, \dots, \mu_h)$, where μ is padded with zeros,

$$\lambda_i > \mu_i \geq 0, \quad \mu_h = 0.$$

Its cells in row i , with rows read from top to bottom, have column indices $\mu_i + 1, \dots, \lambda_i$. Its area, number of rows, and number of columns are denoted n, h, w , respectively. Every column from 1 to $w = \lambda_1$ must contain a cell. The empty diagram is included separately.

The normalization removes horizontal translation. Empty rows are absent by $\lambda_i > \mu_i$. Diagrams are not identified under reflection, transposition, or rotation. We always use λ for the outer partition and μ for the inner partition.

Put

$$\ell_i = \lambda_i - \mu_i > 0, \quad o_i = \lambda_{i+1} - \mu_i \quad (1 \leq i < h). \quad (2.1)$$

The number o_i , when nonnegative, is the number of columns shared by rows i and $i+1$. In particular, $o_i > 0$ means that those rows share a horizontal cell edge. The case $o_i = 0$ is corner contact, not edge connectivity.

Lemma 2.2 (The no-empty-column condition). *A diagram as above has no empty columns if and only if $o_i \geq 0$ for every $1 \leq i < h$.*

Proof. If $\lambda_{i+1} < \mu_i$, every column with index strictly larger than λ_{i+1} and at most μ_i is empty. Rows above the cut start to its right, since their inner-partition parts are at least μ_i ; rows below the cut end to its left, since their outer-partition parts are at most λ_{i+1} .

Conversely, suppose $o_i \geq 0$. The two consecutive row intervals overlap or are adjacent, because $\mu_i + 1 \leq \lambda_{i+1} + 1$. Their union therefore contains no missing integer column. Inducting upwards from the bottom row, which starts in column 1, shows that the union of all row intervals is precisely $\{1, \dots, \lambda_1\}$. Thus no column is empty. $\square$

Proposition 2.3 (Independent overlap coordinates). *Fix positive row lengths $(\ell_1, \dots, \ell_h)$. Normalized diagrams with those row lengths and no empty columns are in bijection with choices*

$$0 \leq o_i \leq \min(\ell_i, \ell_{i+1}) \quad (1 \leq i < h). \quad (2.2)$$

Every choice in these ranges is possible, independently of the others.

Proof. For an existing diagram, the two partition inequalities give

$$o_i \leq \lambda_i - \mu_i = \ell_i, \quad o_i \leq \lambda_{i+1} - \mu_{i+1} = \ell_{i+1},$$

and Lemma 2.2 gives the lower bound.

Conversely, begin with $\mu_h = 0$ and $\lambda_h = \ell_h$. For $i = h-1, h-2, \dots, 1$, define

$$\mu_i = \lambda_{i+1} - o_i, \quad \lambda_i = \mu_i + \ell_i. \quad (2.3)$$

The upper bounds in (2.2) imply

$$\mu_i - \mu_{i+1} = \ell_{i+1} - o_i \geq 0, \quad \lambda_i - \lambda_{i+1} = \ell_i - o_i \geq 0.$$

Thus λ and μ are partitions, all rows are nonempty, and Lemma 2.2 applies. Equations (2.3) also prove uniqueness. $\square$

Corollary 2.4 (Finite general-term sums). *For $n \geq 1$,*

$$a_n = \sum_{h=1}^n \sum_{\substack{\ell_1 + \dots + \ell_h = n \\ \ell_i \geq 1}} \prod_{i=1}^{h-1} (\min(\ell_i, \ell_{i+1}) + 1), \quad (2.4)$$

$$p_n = \sum_{h=1}^n \sum_{\substack{\ell_1 + \dots + \ell_h = n \\ \ell_i \geq 1}} \prod_{i=1}^{h-1} \min(\ell_i, \ell_{i+1}), \quad (2.5)$$

where p_n counts edge-connected diagrams and an empty product equals 1. Moreover, if $a_n(u)$ marks the number of connected components, then

$$a_n(u) = \sum_{h=1}^n \sum_{\substack{\ell_1 + \dots + \ell_h = n \\ \ell_i \geq 1}} u \prod_{i=1}^{h-1} (\min(\ell_i, \ell_{i+1}) + u). \quad (2.6)$$

Proof. Each positive overlap joins its two adjacent rows. Each zero overlap separates two components, and every component consists of consecutive rows. Thus the number of components is one plus the number of zero overlaps. At each boundary there are $\min(\ell_i, \ell_{i+1})$ positive choices and one zero choice; weighting the latter by u , and the first component by u , gives all three formulas. $\square$

The first two sums also explain the composition-based program already recorded in the OEIS entry [1]. They are useful independent checks, but the desired continued fraction is not yet evident from them.

Example 2.5. For row lengths $(2, 2)$ the overlaps $0, 1, 2$ give, respectively,

$$(\lambda, \mu) = ((4, 2), (2, 0)), \quad ((3, 2), (1, 0)), \quad ((2, 2), (0, 0)).$$

The first diagram has two components. The other two are connected. Their widths are $4, 3, 2$, illustrating the general identity

$$w = \lambda_1 = n - \sum_{i=1}^{h-1} o_i. \quad (2.7)$$

This follows by telescoping $\lambda_i = \lambda_{i+1} + \ell_i - o_i$.

2.2 The ordered-sequence construction

Let

$$P(q; s, t) = \sum_{\substack{D \text{ nonempty} \\ D \text{ connected}}} q^{n(D)} s^{h(D)} t^{w(D)}, \quad (2.8)$$

$$\mathcal{A}(q; s, t, u) = \sum_D q^{n(D)} s^{h(D)} t^{w(D)} u^{k(D)}, \quad (2.9)$$

where the empty diagram has weight 1 and $k(D)$ is the number of edge-connected components. Write $P(q) = P(q; 1, 1)$, so $p_0 = 0$.

Proposition 2.6 (Unique concatenation into components). *There is a weight-preserving bijection between all normalized diagrams and finite ordered sequences of nonempty connected normalized diagrams. Consequently,*

$$\mathcal{A}(q; s, t, u) = \frac{1}{1 - uP(q; s, t)}. \quad (2.10)$$

Proof. Cut a diagram between rows i and $i + 1$ whenever $o_i = 0$, then translate each resulting block horizontally to normalize it. Positive overlaps remain within every block, so these blocks are exactly its components. Their order is the order of the row blocks, from top to bottom.

Conversely, concatenate any ordered collection of connected blocks. Retain their internal row lengths and positive overlaps, and insert a zero overlap between each consecutive pair. Proposition 2.3 reconstructs a unique normalized diagram. Geometrically, each upper block is placed northeast of the next block, with their bounding boxes touching at one corner. These two constructions are inverse.

Areas and row counts add. Widths add as well: the inserted overlaps are zero, so (2.7) is additive over blocks. A sequence of k blocks therefore has generating function $u^k P^k$. Summing over $k \geq 0$ gives (2.10); this is legitimate since P has zero constant term in q . $\square$

3 Connected diagrams as weighted Motzkin excursions

3.1 Two boundary paths

Place the diagram in the Cartesian plane, with the bottom row at height zero. A connected diagram of height h and width w is bounded by two north-east lattice paths U and L from $(0, 0)$ to (w, h) . Here U is the upper or northwest boundary, and L is the lower or southeast boundary. The first steps are, respectively, north and east; the last steps are east and north. The two paths have no common point except their endpoints.

For completeness, the partition inequalities make both boundary paths north-east monotone. A meeting at an intermediate horizontal row boundary would say $\lambda_{i+1} = \mu_i$, namely a zero overlap. A shared horizontal boundary segment would force an empty row. Thus positive overlaps and nonempty rows imply that the paths meet only at their endpoints. Conversely, the cells between such a pair of paths have contiguous rows; their left and right endpoints are monotone as rows are read upwards. They therefore form a skew partition diagram, with positive overlaps and no empty rows or columns. This proves the boundary-path bijection.

Both paths have length $L_0 = h + w$. At integer time j , meaning on the line $x + y = j$, let

$$d_j = y_U(j) - y_L(j). \quad (3.1)$$

Then $d_0 = d_{L_0} = 0$, and $d_j \geq 1$ for $1 \leq j < L_0$.

Lemma 3.1 (Area from synchronous separation). *The area of the diagram is*

$$n = \sum_{j=1}^{L_0-1} d_j. \quad (3.2)$$

Proof. Apply the linear map $(x, y) \mapsto (x + y, y)$, whose determinant is 1. The area between the paths becomes the integral of their vertical gap as a function of the new first coordinate. On every interval $[j, j + 1]$ this gap is linear, and its integral is $(d_j + d_{j+1})/2$. Summation gives (3.2), because the endpoint gaps are zero. The original area is the number of unit cells. $\square$

3.2 The four step pairs and their weights

The pairs of simultaneous boundary steps have the following effects:

Upper step	Lower step	Change of gap	Dimension weight
North	East	+1	s
East	North	-1	t
North	North	0, first color	s
East	East	0, second color	t

Dimension weights can be assigned using the upper path alone: its number of north steps is h , and its number of east steps is w .

Delete the forced first up-step and last down-step in the gap process, then lower all remaining gaps by 1. The result is a two-colored Motzkin excursion: a path starting and ending at height zero, staying nonnegative, and using up-steps, down-steps, and two kinds of horizontal steps. Conversely,

the table recovers both boundary words from any such excursion, after the two forced endpoint pairs are restored. There is no ambiguity in either direction.

Give the two removed endpoint steps together the factor stq . For an internal step landing at height b in the Motzkin path, give an additional factor q^{b+1} , together with its dimension weight from the table. Lemma 3.1 proves that the full product is exactly $q^n s^h t^w$. In particular,

$$\begin{aligned} \text{horizontal step at height } j &: (s+t)q^{j+1}, \\ \text{up-step } j \longrightarrow j+1 &: sq^{j+2}, \\ \text{down-step } j+1 \longrightarrow j &: tq^{j+1}. \end{aligned}$$

The two horizontal colors are distinct choices even when $s = t = 1$.

Example 3.2. Take $\lambda = (5, 4, 2)$ and $\mu = (2, 1, 0)$. The row lengths are $(3, 3, 2)$, the overlaps are $(2, 1)$, and $(n, h, w) = (8, 3, 5)$. Its boundary words are

$$U = \text{NENENEEE}, \quad L = \text{EENEENEN}.$$

The interior gaps are $(1, 1, 1, 1, 2, 1, 1)$, whose sum is 8. After removing the forced endpoint steps, the Motzkin heights are $(0, 0, 0, 0, 1, 0, 0)$. The weights give $q^8 s^3 t^5$, as required.

3.3 First-return decomposition and the Jacobi fraction

Let $E_j(q; s, t)$ count excursions beginning and ending at height j and never going below j , with the weights just described. Their absolute height matters because the power of q depends on height.

A nonempty excursion begins either with a horizontal step followed by another excursion at height j , or with an up-step, an excursion at height $j+1$, a down-step, and another excursion at height j . Therefore

$$E_j = 1 + (s+t)q^{j+1}E_j + stq^{2j+3}E_{j+1}E_j.$$

All nonconstant step weights have positive q -degree, so the equation is valid in $\mathbb{Z}[s, t][[q]]$ and yields

$$E_j = \frac{1}{1 - (s+t)q^{j+1} - stq^{2j+3}E_{j+1}}. \quad (3.3)$$

Theorem 3.3 (The connected four-variable building block). *The connected-diagram generating function is*

$$P(q; s, t) = \frac{stq}{1 - (s+t)q - \frac{stq^3}{1 - (s+t)q^2 - \frac{stq^5}{1 - (s+t)q^3 - \frac{stq^7}{1 - (s+t)q^4 - \cdots}}}}. \quad (3.4)$$

Together with (2.10), this counts all normalized diagrams by area, rows, columns, and connected components.

Proof. The bijection gives $P(q; s, t) = stqE_0$. Iterating (3.3) proves (3.4). For any fixed power q^n , only finitely many diagrams and path heights are possible: every step has positive area weight. Hence the successive finite fractions stabilize coefficientwise and define the stated formal series. $\square$

This derivation includes the area statistic explicitly. Using an unweighted Motzkin enumeration here would count a different quantity, essentially semiperimeter, and would not establish the selected formula.

4 Contraction to the conjectured repeated-power fraction

For the area specialization $s = t = 1$, define the paired tails

$$S_j = \frac{1}{1 - \frac{q^j}{1 - q^j S_{j+1}}} \quad (j \geq 1). \quad (4.1)$$

The coefficientwise interpretation is obtained by setting a sufficiently deep tail to 1 and then taking the depth to infinity. Each tail is a unit with constant term 1. Section 7 will give an exact bound on the resulting stabilization.

Lemma 4.1 (Elementary contraction). *If $R_j = S_j - 1$, then*

$$R_j = \frac{q^j}{1 - 2q^j - q^j R_{j+1}} = q^j E_{j-1}(q; 1, 1). \quad (4.2)$$

Proof. Combining the two fractions in (4.1) gives

$$S_j = \frac{1 - q^j S_{j+1}}{1 - q^j - q^j S_{j+1}}.$$

Subtracting 1 and substituting $S_{j+1} = 1 + R_{j+1}$ gives the first identity in (4.2). On putting $R_{j+1} = q^{j+1} E_j$, the denominator becomes $1 - 2q^j - q^{2j+1} E_j$, exactly the denominator of E_{j-1} in (3.3). The finite-depth identities, followed by coefficientwise limits, prove the second equality. $\square$

Proof of Theorem 1.1. The expanded tail S_1 is $C(q)$ from (1.1). By Lemma 4.1 and Theorem 3.3,

$$C(q) - 1 = qE_0(q; 1, 1) = P(q).$$

Proposition 2.6, specialized to $s = t = u = 1$, now gives

$$A(q) = \frac{1}{1 - P(q)} = \frac{1}{2 - C(q)}.$$

Every denominator used has constant term 1. All operations are therefore valid as formal power-series operations, independently of numerical or analytic convergence. $\square$

Corollary 4.2 (Refined repeated-power fraction). *Define*

$$S_j(q; s, t) = \frac{1}{1 - \frac{sq^j}{1 - tq^j S_{j+1}(q; s, t)}}. \quad (4.3)$$

Then

$$P(q; s, t) = t(S_1(q; s, t) - 1), \quad \mathcal{A}(q; s, t, u) = \frac{1}{1 - ut(S_1(q; s, t) - 1)}. \quad (4.4)$$

Proof. Here $R_j = S_j - 1$ satisfies

$$R_j = \frac{sq^j}{1 - (s+t)q^j - tq^j R_{j+1}} = sq^j E_{j-1}(q; s, t).$$

Thus $tR_1 = stqE_0 = P$. Apply (2.10). $\square$

The symmetry in rows and columns is manifest in the Jacobi form (3.4), even though it is not manifest in (4.3). It also follows geometrically by transposing the diagram.

5 A closed formula for fixed rows, columns, and components

The area can be summed out while retaining the dimensions. For each fixed h, w , all diagrams lie in an h -by- w rectangle, so their areas are at most hw . Thus setting $q = 1$ is meaningful coefficientwise as a formal series in s, t , even though it is not a substitution into a convergent area generating function at $q = 1$.

At $q = 1$ the excursion weights no longer depend on height. All E_j become the same series E , with

$$E = 1 + (s+t)E + stE^2.$$

Since $P(1; s, t) = stE$, write $P_*(s, t) = P(1; s, t)$ to obtain

$$P_* = \frac{st}{1 - s - t - P_*} = \frac{1 - s - t - \sqrt{(1 - s - t)^2 - 4st}}{2}, \quad (5.1)$$

where the square root has constant term 1. Consequently

$$\mathcal{A}(1; s, t, u) = \frac{1}{1 - uP_*(s, t)}.$$

Theorem 5.1 (Fixed-dimension general term). *Let $d_{h,w,k}$ count normalized skew diagrams with exactly h rows, w columns, and k edge-connected components, without fixing their area. For $h, w \geq k \geq 1$,*

$$d_{h,w,k} = \frac{k}{h+w-k} \binom{h+w-k}{h} \binom{h+w-k}{w}. \quad (5.2)$$

It is zero if $k \geq 1$ and either $h < k$ or $w < k$. The empty case is $d_{0,0,0} = 1$, and no nonempty diagram has $k = 0$.

Proof. By unique concatenation, $d_{h,w,k} = [s^h t^w] P_*^k$. Introduce a temporary variable z and the unique series

$$Y = \frac{z}{1 - s - t - Y}.$$

Lagrange coefficient extraction gives, for $m \geq k$,

$$\begin{aligned} [z^m] Y^k &= \frac{k}{m} [y^{m-k}] (1 - s - t - y)^{-m} \\ &= \frac{k}{m} \binom{2m-k-1}{m-k} (1 - s - t)^{-(2m-k)}. \end{aligned} \quad (5.3)$$

Substitute $z = st$, so $Y = P_*$. Extracting the remaining powers of s and t gives

$$d_{h,w,k} = k(h+w-k-1)! \sum_{m=k}^{\min(h,w)} \frac{1}{m!(m-k)!(h-m)!(w-m)!}. \quad (5.4)$$

The sum equals

$$\begin{aligned} \frac{1}{h!(w-k)!} \sum_m \binom{h}{m} \binom{w-k}{m-k} &= \frac{1}{h!(w-k)!} \binom{h+w-k}{w} \\ &= \frac{(h+w-k)!}{h!w!(h-k)!(w-k)!}, \end{aligned}$$

by Vandermonde's identity. Substitution in (5.4) is exactly (5.2). The zero cases follow because each nonempty component uses at least one row and one column. $\square$

For $k = 1$, this recovers the usual fixed-dimension parallelogram-polyomino count. For $h = w = 2$, there are three connected diagrams and one diagram with two components. They do not all have the same area, which is why this formula should not be confused with the area coefficients a_n .

6 An independent solution by a ratio of q -series

The boundary-path proof already proves the selected identity. A second algebraic route is valuable both as a check and as an alternative formula for exact coefficient computation. Define

$$\Phi(z; q) = \sum_{r \geq 0} \frac{(-1)^r z^r q^{r(r+1)/2}}{(q; q)_r^2}, \quad (q; q)_r = \prod_{j=1}^r (1 - q^j), \quad (q; q)_0 = 1. \quad (6.1)$$

This is a formal power series in q with polynomial coefficients in z . It is also analytic for $|q| < 1$ and fixed z : on compact subdisks the finite products in the denominator are bounded away from zero, whereas the numerator decays quadratically in the exponent of r .

The connected ratio below is consistent with Bala's treatment and the Delest–Fédou enumeration [7, 8]. We derive it directly from the overlap coordinates, so no continued-fraction identity from those sources is required for this proof.

Theorem 6.1 (The row-refined ratio). *One has*

$$1 + P(q; s, 1) = \frac{\Phi(sq; q)}{\Phi(s; q)}, \quad (6.2)$$

and hence

$$\mathcal{A}(q; s, 1, u) = \frac{\Phi(s; q)}{(1+u)\Phi(s; q) - u\Phi(sq; q)}. \quad (6.3)$$

In particular, if $D(q) = \Phi(1; q)$ and $N(q) = \Phi(q; q)$, then

$$\boxed{A(q) = \frac{D(q)}{2D(q) - N(q)}}. \quad (6.4)$$

Proof. For $i \geq 1$, let X_i count all possible continuations below a previously given row of length i , not counting that row's own weight. The empty continuation is allowed. Appending a row of length j has weight sq^j , and there are $\min(i, j)$ positive overlap choices. Therefore

$$X_i = 1 + s \sum_{j \geq 1} \min(i, j) q^j X_j. \quad (6.5)$$

There is a unique family of formal solutions with constant terms 1: every appended row increases area, so iteration determines one more coefficient at a time. Choosing the initial row separately gives

$$P(q; s, 1) = s \sum_{j \geq 1} q^j X_j = X_1 - 1. \quad (6.6)$$

Set $Z_i = \Phi(sq^i; q)$ for $i \geq 0$. If $c_r = (-1)^r q^{r(r+1)/2} / (q; q)_r^2$, then $c_r(1 - q^r)^2 = -q^r c_{r-1}$ for $r \geq 1$. Comparing coefficients of s^r now gives

$$Z_{i+1} - (2 - sq^i)Z_i + Z_{i-1} = 0 \quad (i \geq 1). \quad (6.7)$$

Let $\Delta_i = Z_i - Z_{i-1}$. Both $Z_i \rightarrow 1$ and $\Delta_i \rightarrow 0$ coefficientwise as $i \rightarrow \infty$. Equation (6.7) says

$$\Delta_i - \Delta_{i+1} = sq^i Z_i.$$

Summing first over $i \geq a$, and then over $1 \leq a \leq k$, gives

$$\begin{aligned} \Delta_a &= s \sum_{j \geq a} q^j Z_j, \\ Z_k - Z_0 &= s \sum_{j \geq 1} \min(k, j) q^j Z_j. \end{aligned}$$

Every sum is coefficientwise finite. The series $Z_0 = \Phi(s; q)$ has constant term 1, so division by it is valid. The family Z_k/Z_0 solves (6.5), and uniqueness implies $X_k = Z_k/Z_0$. Equation (6.6) proves (6.2). Finally apply the component identity (2.10) and simplify. $\square$

Expanding the two series in (6.4) explicitly gives

$$D(q) = \sum_{r \geq 0} \frac{(-1)^r q^{r(r+1)/2}}{(q; q)_r^2}, \quad N(q) = \sum_{r \geq 0} \frac{(-1)^r q^{r(r+3)/2}}{(q; q)_r^2}.$$

Thus the selected continued fraction and a ratio of explicit infinite sums have now been derived from two different encodings of the same diagrams.

6.1 A finite transfer matrix and a different continued fraction

This construction will also supply an exact numerical certificate. Restrict every row length to be at most H , and let $P_H(q)$ count the connected diagrams satisfying that restriction. Set

$$M_H = (\min(i, j))_{1 \leq i, j \leq H}, \quad D_H = \text{diag}(q, q^2, \dots, q^H),$$

and let $\mathbf{e}$ be the all-ones column vector. The transfer equation becomes

$$\mathbf{X} = \mathbf{e} + M_H D_H \mathbf{X}, \quad P_H = \mathbf{e}^\top D_H \mathbf{X}. \quad (6.8)$$

The matrix M_H is $LL^\top$, where L is lower triangular with all entries on and below its diagonal equal to 1. Its inverse is

$$B_H = M_H^{-1} = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 1 \end{pmatrix} \quad (H \geq 2), \quad B_1 = (1). \quad (6.9)$$

Since $B_H \mathbf{e}$ is the first standard basis vector, and $X_1 = 1 + P_H$, it follows that

$$1 + P_H(q) = (B_H - D_H)_{11}^{-1}. \quad (6.10)$$

Successive Schur complements give the entirely finite fraction

$$1 + P_H(q) = \frac{1}{2 - q - \frac{1}{2 - q^2 - \frac{1}{\ddots - \frac{1}{2 - q^{H-1} - \frac{1}{1 - q^H}}}}} \quad (H \geq 2), \quad (6.11)$$

with $1 + P_1(q) = 1/(1 - q)$. Notice the final denominator $1 - q^H$, not $2 - q^H$. This boundary condition matters in both exact computation and error bounds.

7 Sharp truncation and an exact coefficient algorithm

7.1 The first omitted term is a square

Let $C^{(m)}(q)$ be the paired fraction obtained by setting $S_{m+1}^{(m)} = 1$ and applying (4.1) for $j = m, m-1, \dots, 1$. Put

$$A^{(m)}(q) = \frac{1}{2 - C^{(m)}(q)}.$$

For $m = 0$ this means $C^{(0)} = A^{(0)} = 1$.

Theorem 7.1 (Exact truncation order). *For every integer $m \geq 0$,*

$$C(q) - C^{(m)}(q) = q^{(m+1)^2} + O(q^{(m+1)^2+1}), \quad (7.1)$$

$$A(q) - A^{(m)}(q) = q^{(m+1)^2} + O(q^{(m+1)^2+1}). \quad (7.2)$$

Here $O(q^r)$ means a formal series divisible by q^r . Consequently $m = \lfloor \sqrt{N} \rfloor$ pairs suffice to compute every coefficient through q^N .

Proof. Write the paired-tail map as

$$\phi_j(z) = \frac{1 - q^j z}{1 - q^j - q^j z}.$$

For any two formal units a, b with constant term 1, direct subtraction gives

$$\phi_j(a) - \phi_j(b) = \frac{q^{2j}(a - b)}{(1 - q^j - q^j a)(1 - q^j - q^j b)}. \quad (7.3)$$

At the first omitted tail,

$$S_{m+1} - 1 = q^{m+1} + O(q^{m+2}).$$

Applying (7.3) for $j = m, m-1, \dots, 1$ increases the valuation by $2 \sum_{j=1}^m j = m(m+1)$ and leaves the leading coefficient equal to 1. This proves (7.1), since $m(m+1) + (m+1) = (m+1)^2$. It also proves directly that the finite fractions stabilize coefficientwise.

Finally,

$$\frac{1}{2 - C} - \frac{1}{2 - C^{(m)}} = \frac{C - C^{(m)}}{(2 - C)(2 - C^{(m)})}.$$

Both denominator factors have constant term 1, so the same valuation and leading coefficient hold for $A - A^{(m)}$. Since $(\lfloor \sqrt{N} \rfloor + 1)^2 > N$, the stated depth is sufficient. $\square$

There is a geometric explanation. Contracting a fraction of depth m allows Motzkin heights at most $m-1$, or boundary gaps at most m . The least-area omitted excursion rises to height m and immediately descends, with no horizontal steps. Its diagram is the $(m+1)$ -by- $(m+1)$ square, with area $(m+1)^2$. It is unique at that least area, explaining the leading coefficient 1. Additional components would increase the area, so the same first discrepancy holds for all diagrams.

For $N = 1000$, only 31 pairs, that is, 62 individual partial numerators, are needed. The first missing term has degree $32^2 = 1024$. This statement is exact, not an empirical rule about rapid numerical convergence.

7.2 Continuants avoid repeated series inversion

Write a finite tail as U/V , with $U(0) = V(0) = 1$. Then

$$\phi_j(U/V) = \frac{V - q^j U}{V - q^j V - q^j U}.$$

Thus the finite polynomials U, V for $C^{(m)} = U/V$ can be computed by starting with $U = V = 1$ and performing the simultaneous updates

$$\begin{aligned} U_{\text{new}} &= V - q^j U, \\ V_{\text{new}} &= V - q^j V - q^j U, \end{aligned} \quad j = m, m-1, \dots, 1. \quad (7.4)$$

All polynomials may be truncated after degree N . At the end,

$$A^{(m)} = \frac{V}{2V - U}. \quad (7.5)$$

Only one series division is needed to obtain A .

If $B(q) = \sum b_j q^j = 2V - U$ and $V(q) = \sum v_j q^j$, then $b_0 = 1$ and

$$a_0 = 1, \quad a_n = v_n - \sum_{j=1}^n b_j a_{n-j} \quad (1 \leq n \leq N). \quad (7.6)$$

The shift-and-subtract updates cost $O(N\sqrt{N})$ integer arithmetic operations. Ordinary dense division costs $O(N^2)$ such operations. The continuant method uses $O(N)$ integer storage. These are arithmetic-operation bounds, not bit-complexity bounds; the integers grow with N .

The included implementation also evaluates the overlap sums by a different dynamic program. Let $f_{n,i}$ count diagrams of area n whose last row has length i . Initialize $f_{i,i} = 1$, and append a row of length j with multiplier $\min(i, j) + \varepsilon$, where $\varepsilon = 0$ for connected shapes and $\varepsilon = 1$ for all shapes. The transition sum can be written

$$\sum_i f_{n,i}(\min(i, j) + \varepsilon) = \sum_{i \leq j} i f_{n,i} + j \sum_{i > j} f_{n,i} + \varepsilon \sum_i f_{n,i}. \quad (7.7)$$

Prefix sums give an $O(N^2)$ -operation dynamic program, using $O(N^2)$ storage in the supplied straightforward implementation. It never evaluates a continued fraction.

8 Rigorous exponential asymptotics

8.1 Analytic control from the row-transfer kernel

The formal proofs imply exact enumeration. To derive asymptotics, we now establish an analytic disk large enough to contain the dominant pole. For $s = 1$, regard the transfer kernel

$$K(q)_{ij} = \min(i, j)q^j \quad (i, j \geq 1)$$

as an operator on the Banach space of bounded sequences, with the supremum norm. For $|q| \leq r < 1$,

$$\|K(q)\| \leq \sum_{j \geq 1} jr^j = \frac{r}{(1-r)^2} =: \eta(r). \quad (8.1)$$

The operator-valued series defining K is analytic in that disk. In particular, if

$$r < r_0 := \frac{3 - \sqrt{5}}{2} = 0.381966\dots, \quad (8.2)$$

then $\eta(r) < 1$, so

$$\mathbf{X} = (I - K)^{-1} \mathbf{e} = \sum_{m \geq 0} K^m \mathbf{e}$$

converges uniformly in operator norm. The functional $\mathbf{f}(q)^\top \mathbf{X} = \sum_{j \geq 1} q^j X_j$ has norm bounded by $r/(1-r)$. Hence $P(q)$ is analytic for $|q| < r_0$.

Lemma 8.1 (Existence of a positive singularity before $1/3$). *There is a unique $\rho \in (0, 1/3)$ such that $P(\rho) = 1$, and $P'(\rho) > 0$.*

Proof. The coefficients of P are nonnegative, and $p_1 = 1$. Thus P is strictly increasing on $(0, r_0)$, with $P(0) = 0$ and positive derivative. Its first seven terms are

$$q + 2q^2 + 4q^3 + 9q^4 + 20q^5 + 46q^6 + 105q^7.$$

At $q = 1/3$ their sum is $245/243 > 1$. Analyticity at $1/3$, continuity, and strict monotonicity now prove the assertion. $\square$

Theorem 8.2 (The asymptotic formula). *Let ρ be as in Lemma 8.1. There is an $R > \rho$ such that*

$$a_n = \kappa \rho^{-n} + O(R^{-n}), \quad \kappa = \frac{1}{\rho P'(\rho)} > 0. \quad (8.3)$$

In particular, $\lim_{n \rightarrow \infty} a_n^{1/n} = 1/\rho$, and $a_{n+1}/a_n \rightarrow 1/\rho$.

Proof. For $|q| < \rho$, $|P(q)| \leq P(|q|) < 1$, so $A = 1/(1 - P)$ has no pole. If $|q| = \rho$ and $P(q) = 1$, equality holds in the triangle inequality for $\sum_{n \geq 1} p_n q^n$. The terms of degrees 1 and 2 have positive coefficients; their phases must agree. Writing $q = \rho e^{i\theta}$ gives $e^{i\theta} = e^{2i\theta}$, hence $q = \rho$. Thus ρ is the only zero of $1 - P$ on its circle.

At ρ , the derivative $P'(\rho)$ is positive, so the pole is simple. Because P is analytic in a larger disk and analytic zeros are isolated, there are numbers $\rho < R < R' < r_0$ for which $1 - P$ has no other zero in $|q| \leq R'$. Its principal part at ρ is

$$\frac{1}{\rho P'(\rho)} \frac{1}{1 - q/\rho}.$$

Subtracting this function from A removes the pole. The difference is analytic on $|q| < R'$, so Cauchy's coefficient bound on $|q| = R$ is $O(R^{-n})$. This proves (8.3). The two stated limits follow because $\kappa > 0$. $\square$

For reference, $D(q) = \Phi(1; q)$ does not vanish in $|q| < r_0$. Indeed, for fixed q there the bounded sequence $Z_i = \Phi(q^i; q)$ satisfies $\mathbf{Z} = D\mathbf{e} + K\mathbf{Z}$, by the proof of Theorem 6.1. If $D = 0$, invertibility of $I - K$ forces $\mathbf{Z} = 0$, contradicting $Z_i \rightarrow 1$. Therefore the equation for ρ can equivalently be written

$$2\Phi(1; \rho) - \Phi(\rho; \rho) = 0, \quad (8.4)$$

with a nonzero numerator in (6.4).

8.2 A rational certificate for the dominant root

High-precision evaluation suggests

$$\rho \approx 0.3195967180593874655186028919827139236, \quad (8.5)$$

$$\rho^{-1} \approx 3.128943269730886252277447995387754161, \quad (8.6)$$

$$\kappa \approx 0.2893214639716513230848453480726559061. \quad (8.7)$$

These displayed extended decimals are numerical estimates. The following strict interval is separately certified using exact rational arithmetic:

$$\boxed{0.31959671805938746551 < \rho < 0.31959671805938746552.} \quad (8.8)$$

Here is the complete bound behind that certificate. For real $0 < q < r_0$, define

$$\tau_H(q) = \sum_{j>H} q^j = \frac{q^{H+1}}{1 - q}, \quad (8.9)$$

$$\delta_H(q) = \sum_{j>H} j q^j = \frac{q^{H+1}((H+1) - Hq)}{(1 - q)^2}. \quad (8.10)$$

Proposition 8.3 (An explicit bounded-row error). *For $0 < q < r_0$,*

$$0 \leq P(q) - P_H(q) \leq \frac{\tau_H(q)}{1 - \eta(q)} + \frac{q}{1 - q} \frac{\delta_H(q)}{(1 - \eta(q))^2} =: \mathcal{E}_H(q). \quad (8.11)$$

Proof. Extend the finite kernel to bounded infinite sequences by suppressing all columns with index $j > H$; denote this operator by K_H . Its first H coordinates satisfy the finite system (6.8). The operator norms obey

$$\|K_H\| \leq \eta(q), \quad \|K - K_H\| \leq \delta_H(q).$$

Let $\mathbf{f}$ be the functional with entries q^j , and $\mathbf{f}_H$ its truncation. Then

$$\|\mathbf{f} - \mathbf{f}_H\| = \tau_H(q), \quad \|\mathbf{f}_H\| \leq \frac{q}{1 - q}.$$

Using the resolvent identity,

$$(I - K)^{-1} - (I - K_H)^{-1} = (I - K)^{-1}(K - K_H)(I - K_H)^{-1},$$

and $\|\mathbf{e}\| = 1$, we obtain the upper bound in (8.11). The lower bound also follows directly because P_H counts a subset of the diagrams counted by P . $\square$

Let q_- and q_+ denote the two rational endpoints in (8.8). The script `verify.py` evaluates (6.11) with $H = 55$ using Python's exact rational type and checks

$$P_{55}(q_-) + \mathcal{E}_{55}(q_-) < 1, \quad P_{55}(q_+) > 1. \quad (8.12)$$

The error allowance at the lower endpoint is about 7.446×10^{-26} . No floating-point decision is used in (8.12). Proposition 8.3, combined with monotonicity of P , proves (8.8). The interval has width 10^{-20} ; this statement should not be confused with certification of every additional decimal printed in (8.5)–(8.7).

9 The number of connected components in a large diagram

The component marker gives more than a counting refinement. Choose a diagram uniformly among the a_n diagrams of area n , for $n \geq 1$, and let K_n be its number of edge-connected components. By (2.10),

$$\mathbb{E}[u^{K_n}] = \frac{[q^n](1 - uP(q))^{-1}}{a_n}. \quad (9.1)$$

Theorem 9.1 (Linear mean, linear variance, and a Gaussian limit). *Define*

$$\mu = \frac{1}{\rho P'(\rho)}, \quad \sigma^2 = \mu^2 - \mu + \frac{P''(\rho)}{\rho P'(\rho)^3}. \quad (9.2)$$

Then $\sigma^2 > 0$,

$$\mathbb{E}[K_n] = \mu n + O(1), \quad \text{Var}(K_n) = \sigma^2 n + O(1), \quad (9.3)$$

and

$$\frac{K_n - \mu n}{\sigma \sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1). \quad (9.4)$$

Numerically, $\mu = \kappa \approx 0.2893214639716513231$ and $\sigma^2 \approx 0.2382004452790812020$.

Proof. The implicit function theorem gives an analytic function $\rho(u)$ near $u = 1$ such that

$$uP(\rho(u)) = 1, \quad \rho(1) = \rho.$$

Choose a circle $|q| = R > \rho$ as in Theorem 8.2. For complex u in a sufficiently small neighborhood of 1, continuity and the argument principle show that $1 - uP(q)$ has just the one simple zero $\rho(u)$ inside that circle and no zero on it. Subtracting the moving principal part and applying Cauchy's bound uniformly gives

$$[q^n] \frac{1}{1 - uP(q)} = \frac{\rho(u)^{-n}}{u\rho(u)P'(\rho(u))} + O(R^{-n}). \quad (9.5)$$

The leading factor is analytic and nonzero near $u = 1$.

Set $r(t) = \rho(e^t)$ and $L(t) = \log(\rho/r(t))$, taking local analytic branches. Equation (9.1) and the uniform estimate imply

$$\log \mathbb{E}[e^{tK_n}] = nL(t) + G(t) + O(\theta^n), \quad 0 < \theta < 1, \quad (9.6)$$

locally uniformly in complex t , for an analytic G independent of n . Differentiate $P(r(t)) = e^{-t}$ at zero to obtain

$$r'(0) = -\frac{1}{P'(\rho)}, \quad r''(0) = \frac{1}{P'(\rho)} - \frac{P''(\rho)}{P'(\rho)^3}.$$

It follows that $L'(0) = \mu$ and $L''(0) = \sigma^2$, proving (9.3) by differentiation of the logarithmic moment generating function.

To verify strict positivity, set $w_j = p_j \rho^j$. Since $P(\rho) = 1$, these numbers form a probability distribution on positive integers. Its mean m and variance v satisfy

$$m = \rho P'(\rho), \quad v = \rho^2 P''(\rho) + \rho P'(\rho) - (\rho P'(\rho))^2.$$

A direct simplification gives $\sigma^2 = v/m^3$. Both w_1 and w_2 are positive, so this distribution is not concentrated at one point; hence $v > 0$.

Finally, substitute $t = iz/(\sigma\sqrt{n})$ in (9.6) for fixed real z and subtract the centering term $iz\mu\sqrt{n}/\sigma$. Taylor expansion gives a limit of $-z^2/2$ for the logarithm of the characteristic function. Its exponential therefore tends to $e^{-z^2/2}$, proving (9.4) by the continuity theorem for characteristic functions. $\square$

The equality between the mean-component density μ and the leading coefficient κ in the area asymptotic is a consequence of the same simple-pole calculation, not a numerical coincidence. These consequences are derived here without a claim that their general analytic mechanism is new.

10 Exact checks, numerical checks, and reproducibility

10.1 Initial values and component polynomials

Table 2 compares the two area sequences obtained from the proof. The connected sequence is A006958, with its constant term set to zero; the all-diagram sequence is A225114, including the empty object [6, 1].

Area n	Connected diagrams p_n	All diagrams a_n
0	0	1
1	1	1
2	2	3
3	4	9
4	9	28
5	20	87
6	46	272
7	105	850
8	242	2659
9	557	8318
10	1285	26025
11	2964	81427
12	6842	254777

Table 2: Initial exact coefficients.

For example, direct overlap enumeration gives

$$\begin{aligned}
a_1(u) &= u, \\
a_2(u) &= 2u + u^2, \\
a_3(u) &= 4u + 4u^2 + u^3, \\
a_4(u) &= 9u + 12u^2 + 6u^3 + u^4, \\
a_5(u) &= 20u + 34u^2 + 24u^3 + 8u^4 + u^5, \\
a_6(u) &= 46u + 92u^2 + 83u^3 + 40u^4 + 10u^5 + u^6.
\end{aligned}$$

Their coefficients agree with $[q^n]P(q)^k$ for each component count k . Setting $u = 1$ recovers the corresponding values in the table.

10.2 The exact verification suite

The following checks were executed successfully by `python3 verify.py -limit 1000`. The summary is preserved in `verification_results.json`.

1. The recorded candidate-list digest and random draw were replayed without creating a new draw.
2. All 31 initial terms displayed in A225114 and all 32 positive-area terms displayed in A006958 agreed with the computed values.
3. Independent row-overlap dynamic programs, the paired continued fraction, the explicit q -series ratio, and the ordered-component convolution agreed through area 1000.
4. The exact first omitted coefficient in Theorem 7.1 was checked at every depth from 0 to 12.
5. Full component polynomials were independently enumerated from compositions through area 13.
6. The four-variable refinement was checked through area 20 at all 27 triples $(s, t, u) \in \{0, 1, 2\}^3$, against direct weighted row placements. These are specialized tests, not a claim of symbolic verification at all four-variable coefficients through area 1000.

7. Direct enumeration of the two boundary paths agreed with the connected area counts through area 10. Since $n \geq h + w - 1$ by (3.2), enumerating boundary lengths through 11 covers every diagram in that range.
8. The dimension formula (5.2) agreed with independent boundary-path enumeration and concatenation for every $h, w \leq 5$ and $1 \leq k \leq 5$.
9. The dominant-root inequalities (8.12) were verified with exact rational arithmetic, and the seven-term lower bound $P(1/3) > 245/243$ was checked.

The program reported **ALL EXACT CHECKS PASSED**. These finite checks are designed to expose indexing, weighting, truncation, and implementation errors. They are not used as a replacement for any all-orders proof in Sections 2–9.

10.3 Numerical diagnostics for the asymptotics

The optional script `asymptotics.py` uses `mpmath` at high precision to estimate ρ , $P'(\rho)$, and $P''(\rho)$. It compares continued-fraction depths 40 and 60 as a stability check. Agreement of those evaluations is a numerical diagnostic, not the rational certificate; only (8.8) is certified in that way.

The moments can also be recovered from integer coefficient arrays, using

$$\mathbb{E}[K_n] = \frac{[q^n](A^2 - A)}{a_n}, \quad \mathbb{E}[K_n(K_n - 1)] = \frac{2[q^n](A^3 - 2A^2 + A)}{a_n}. \quad (10.1)$$

These follow by differentiating $(1 - uP)^{-1}$ and using $P = 1 - 1/A$. Table 3 compares the resulting statistics with their limits.

n	$a_n \rho^n$	$\mathbb{E}[K_n]/n$	$\text{Var}(K_n)/n$
10	0.289343577662	0.371627281460	0.209156420607
20	0.289321476436	0.330486811475	0.223648203436
50	0.289321463972	0.305787608032	0.232379523182
100	0.289321463972	0.297554536002	0.235289984230
250	0.289321463972	0.292614692784	0.237036260860
500	0.289321463972	0.290968078378	0.237618353069
1000	0.289321463972	0.290144771175	0.237909399174
Limit	0.289321463972	0.289321463972	0.238200445279

Table 3: Numerical diagnostics from exact area and component counts, with high-precision estimates of the asymptotic constants.

10.4 Contents and scope of the accompanying archive

The archive contains the editable $\text{\LaTeX}$ source and its compiled PDF, the frozen candidate list and selection record, the standard-library exact verification program, the optional numerical program, and CSV/JSON outputs. The coefficient table contains $n = 0, 1, \dots, 1000$. The separate component table contains the nonzero entries for $n \leq 13$. The README gives build and execution commands and distinguishes rigorous inequalities from numerical estimates.

No proof assistant or external referee has certified this article, and no edit has been submitted to OEIS as part of this work. The all- n result rests on the explicit mathematical arguments above; the code provides independent reproducible checks of their consequences.

11 Conclusion

The generating function selected from the OEIS shortlist is correct. Its structure has a direct combinatorial explanation: zero row overlaps cut a diagram into an ordered sequence of connected pieces; each connected piece is encoded by two separated monotone boundaries; their separation is a weighted Motzkin excursion. First-return decomposition produces the Jacobi fraction, and elementary contraction produces exactly the repeated powers in the selected expression.

The same framework yields the full refinement $\mathcal{A}(q; s, t, u) = (1 - uP(q; s, t))^{-1}$, the dimension formula (5.2), the independent q -series ratio (6.4), the sharp truncation degree $(m + 1)^2$, and the rigorous asymptotic (8.3). The prior sketch is explicitly credited; the deliverable is a complete proof and reproducible elaboration of a formula still labeled conjectural in its OEIS entry, not a priority claim.

A The unselected candidate formulas and draw record

The following formulas were present in the frozen list. They are recorded to make the selection auditable; this article makes no assertion that they have been proved or refuted here.

For A244475, with the power-series indexing beginning at $n = 3$, the conjectured generating function is [2]

$$-\frac{q^3 Q(q)}{q^2 + q - 1}, \quad Q(q) = q^{14} + q^{13} + q^{12} + 2q^{11} + 3q^{10} + 5q^9 + 8q^8 + q^7 + 3q^6 + 3q^5 + 2q^4 + 4q^3 + 5q^2 + 2q + 1. \quad (\text{A.1})$$

The OEIS attribution is Alois P. Heinz, 20 June 2022.

For A381190, with indexing beginning at $n = 3$, the conjectured generating function is [3]

$$-\frac{2q^3(4q^5 + 8q^4 + 4q^3 - 9q^2 - 8q - 3)}{(q^3 + q^2 - 1)^2}, \quad (\text{A.2})$$

attributed to Joerg Arndt, 7 January 2026.

For A391632, with indexing beginning at $n = 1$, the conjectured generating function is [4]

$$\frac{q(4 - 19q + 33q^2 - 40q^3 + 8q^4)}{1 - 10q + 29q^2 - 32q^3 + 8q^4}, \quad (\text{A.3})$$

attributed to Andrew Howroyd, 12 January 2026.

The SHA-256 digest of the exact UTF-8 bytes of `candidates.json` is

ea6cca457acda9a1fbd92f88aa09cd9250053f0f467d75e0f7323aabc7c82d37

The RNG seed is given in Section 1; the recorded output is 0. The exact verification program replays this draw without altering the record. The original selection script deliberately refuses to run when `selection.json` already exists, to avoid silently replacing the recorded selection.

B A minimal reproduction protocol

Compile the article twice with a standard L^AT_EX installation:

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

Run the exact checks with Python 3.10 or later:

```
python3 verify.py --limit 1000
```

This step uses only the Python standard library. For the optional numerical tables, install the separately listed dependency and run:

```
python3 -m pip install -r requirements.txt
python3 asymptotics.py
```

The numerical script computes its own coefficient arrays through area 1000 using the continuant routine. Running the exact script with a different limit overwrites its CSV coefficient output at that limit; use 1000 to reproduce the supplied exact table. Neither command performs a new random selection.

References

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- [8] M.-P. Delest and J.-M. Fédou, *Enumeration of skew Ferrers diagrams*, Discrete Mathematics **112** (1993), 65–79. [https://doi.org/10.1016/0012-365X\(93\)90224-H](https://doi.org/10.1016/0012-365X(93)90224-H). Historical background; the identities used here are independently derived in the article.